Computer Vision Term Project

CS 423/523 – Fall 2026

1. Overview

The term project gives you an opportunity to apply the subjects learned in class to a computer vision problem of your choice. In the CS 423/523 project students are expected to clearly formulate a problem, compare appropriate methods on standard data, and analyse the results they will obtain critically. Students may choose any topic relevant to the course content. Projects are evaluated on the basis of the soundness of the problem formulation, the rigour of the evaluation, and the quality of the analysis. These criteria will be checked based on your report and the the presentation you will give.

Sample project sources

- https://www.v7labs.com/blog/computer-vision-project-ideas
- http://cs231n.stanford.edu/2017/reports.html
- https://www.skyfilabs.com/blog/best-computer-vision-projects-for-engineering-students
- https://data-flair.training/blogs/computer-vision-project-ideas/

2. Project Tracks

Students may choose one of the following three tracks and state it in their proposal.

Track A – Application
Apply computer vision to a real-world problem, ideally related to your background or interests (e.g., biomedical, engineering, physics, robotics, agriculture, etc.). The main body of the work in this track is problem formulation, dataset (collected, curated, or carefully selected and cleaned), and a rigorous comparison of existing techniques along with proper evaluations. Adapting or fine-tuning existing models for the chosen task is expected.

Track B – Algorithm
Design and implement a new algorithm, or a meaningful improvement of an existing one, using classical computer vision, machine learning, or deep learning, and evaluate it on a standard vision task (e.g., recognition, detection, segmentation, tracking, 3D vision, image-to-text, text-to-image, etc.). The core body of the work in this track is a clearly stated technical idea, its justification, and ablation studies showing why and how it works. Small modifications that are not motivated or analysed will not qualify.

Track C – Reproduction and Analysis
Reproduce the main results of a recent (past 5 years maximum) paper (preferably with public code or data) published in a prestigious venue and then go beyond it: test its robustness to distribution drift or corruptions, analyse failure modes, examine possible bias, study sensitivity to hyperparameters or data size, or evaluate on a new dataset. The core of the work in this track is a *faithful* reproduction, a clear report of any discrepancies, and an original analysis that adds insight beyond what was initially reported in the corresponding paper.

What each track emphasizes

Track	Primary emphasis	Minimum expectation
A	Problem formulation, data quality, evaluation rigour	At least three methods must be compared under a common protocol, with error analysis
B	Technical idea, novelty, ablation studies	A working implementation of the proposed idea, compared with strong recent baselines, with ablations
C	Reproduction fidelity, depth of analysis	Reported performances/numbers reproduced (or discrepancies explained) plus at least one substantive extension

Expectations by course section

CS 423 teams may choose any of the three tracks. CS 523 projects are additionally expected to include a methodological contribution or a notably deeper analysis (e.g., Track B, or Track A/C with a clear original element beyond applying or re-running existing methods).

3. Requirements Common to All Tracks

- **Baselines.** Compare against appropriate and strong baselines, including *at least one simple or classical method* (non-deep) where a meaningful one exists (e.g., hand-crafted features with a linear or kernel classifier, template matching, a pretrained feature extractor with a linear head). If you argue that no meaningful classical baseline exists for your task, justify this in the report.
- **Data splits and leakage.** Use proper training, validation, and test splits. Tune hyperparameters on validation data only, and use the test set only for the final evaluation. Avoid data/label leakage (e.g., near-duplicate images or frames from the same video or patient across splits).
- **Variability.** Report results over multiple runs or random seeds (report mean and standard deviation whenever applicable) where training is stochastic, or otherwise state clearly why this was not feasible.
- **Metrics.** Use standard metrics appropriate for the problem and the data distribution (e.g., per-class metrics or AP/mIoU rather than accuracy alone on imbalanced data), and explain your choices.
- **Error analysis.** Include qualitative samples and a discussion of typical failure cases.
- **Reproducibility.** Provide code, configuration, and instructions that allow your main results to be re-run (see Section 8).
- **Data ethics and licensing.** Respect dataset licenses and privacy. If you collect your own data involving people, discuss consent and anonymization. Discuss potential bias or misuse where relevant.
- **Deep learning projects.** If you use deep learning, the classical-baseline comparison above is required, and you must discuss the computational budget used (hardware and approximate training time).

4. Teams

Course section	Team size
CS 423 (undergraduate)	3 students (in exceptional cases, 2 only with instructor approval)
CS 523 (graduate)	1 or 2 students

Larger teams are expected to deliver correspondingly more substantial project work. Only one team member will submit the project on behalf of the team. The report must include a short *contribution statement* listing what each member did, and individual grades may be adjusted accordingly.

5. Stages and Deadlines

Stage	Deliverable	Due
1	Your group preference, if any, via email to course TAs	15 October 2026
2	Confirmed project proposal (PDF)	1 November 2026
3	Final report (PDF), code and supplementary material (ZIP/PDF)	18 December 2026
4	Project presentation	18-25 December 2026

- Once you decided on your team members, you must email the course TAs (Abdelaziz Hussein (abdelaziz.hussein@ozu.edu.tr) and Yücel Aytaç Akgün (aytac.akgun@ozu.edu.tr)) the names and student numbers of your group members.
- Once you have decided on your project topic and prepared your proposal, must check with the course TA (Abdelaziz Hussein (abdelaziz.hussein@ozu.edu.tr) and Yücel Aytaç Akgün (aytac.akgun@ozu.edu.tr)) and seek confirmation of the appropriateness of your project topic. Similar projects are NOT allowed; the project topics will be approved on a first-come-first-approved basis.
- All materials are submitted through the course LMS. Any submission material turned in late will be penalised and will incur a reduction of 25% in the final score, for each day (or part thereof) it is late. We follow no deadline extension policy.

6. Project Proposal

The proposal is **1–2 pages** (PDF) and must contain:

1. **Title, team members, and track** (A, B, or C).
2. **Abstract.** One short paragraph describing the scope of the project and what you plan to accomplish.
3. **Introduction.** Problem definition and motivation.
4. **Problem statement and methodology.**
   - The outline of the technical methods to be applied or compared, including baselines.
   - The dataset(s) to be used (source, size, license, splits).
   - The expected results, with reference to the current state of the art.
   - The evaluation protocol: metrics and success criteria.

5. **Risks and fallback plan.** The main risks (data availability, compute, reproducibility) and what you will do if the primary plan fails.
6. **Project schedule.** A list of subtasks with, for each, the planned approach, a deadline, and the responsible team member. Allocate time for the literature search, implementation and experiments, the final report, and the presentation.
7. **References.** At least **5** relevant papers, documents, or web links, each with a complete citation (authors, title, venue, and URL where appropriate). Track C proposals must identify the paper whose results shall be reproduced.

7. Final Report

The final report must follow the **IEEE journal article format** and use the **IEEE citation style**. Templates (both LaTeX and Word) are available from the IEEE Template Selector. If you prefer to prepare your report in LaTeX, you can use the official IEEE journal article templates on Overleaf (e.g., the "IEEE Journal Paper Template"), which can be opened and edited online in your browser without installing anything locally; see Overleaf for a free account. The length is **6–10 pages** including references, excluding supplementary material. Every figure and table must have a caption and be discussed in the main text. Your final project report is expected to be organised as follows.

1. **Title and authors.**
2. **Abstract** (at most 200 words): problem, approach, and key results.
3. **Introduction.** The problem, why it matters, an overview of your results, and a clear statement of your contributions.
4. **Related Work.** The state of the art and published work related to your project; how your approach is similar to or differs from it.
5. **Data.** Type, origin, size, splits, and any preprocessing, filtering, or special treatment.
6. **Methods.** Your approach, possibly in several subsections (one per technique). Justify the choices and mention the alternatives considered. When comparing techniques, summarize their advantages and disadvantages. For Track C, describe the original method and how your reproduction differs from it.
7. **Experimental Resultsu:** The experimental setup (implementation details, hyperparameters, etc.), the results and comparison with baselines and the state of the art, ablation or sensitivity studies, visualizations that provide insight into the behaviour of the method, and a failure-case analysis. Report statistical variability as required in Section 3.
8. **Discussion and Conclusion.** Key findings and lessons learned, limitations, and ideas for future work.
9. **Contribution statement** of each team member.
10. **Acknowledgement** of AI tool use (Section 10).
11. **References.** Every reference must be cited in the main text. Include papers, web pages, GitHub repositories, and any other code or data used.

8. Code and Supplementary Material

- Submit your **source code** as a single `.zip` file. Include a `README` file precisely describing the environment, how to obtain the data, and the commands needed to reproduce the main

results reported in your project. Do not include large datasets or model weights, instead provide links.

- **Supplementary material** (not counted toward the page limit) may include videos, interactive visualizations, demos, and additional results. Submit it as a separate PDF or ZIP file.
- **Code attribution.** Any code you used as part of your project must be cited in the main body of the report. This includes assignment code, online examples, open-source projects, and GitHub implementations. A footnote or a full bibliography entry is acceptable. Clearly distinguish what you wrote from what you reused.

9. Final Presentation

Each team will be given **10–15 minute** for an oral presentation of their project work, including a demonstration where appropriate, followed by a short Q&A. Each team member must contribute to the presentation in a meaningful way..

The presentation content, its format and layout, and the way that you present it will affect your grade. Prepare your slides neatly and properly. It should contain roughly 10–15 slides with reasonable content (only present the most important and interesting parts). Do not copy and paste any text/equation/table from a paper (if necessary, type them). If you need to use a figure (or an image) of a paper, take it but give a credit to the source.

Suggested outline

1. Project title and team members
2. Problem definition and motivation (1–2 slides)
3. Project overview: goal, subtasks, and who did what
4. Literature review (1–2 slides)
5. Method (5–6 slides)
6. Experimental results (3–4 slides): datasets, metrics, results, comparison with baselines and the state of the art
7. Demo, if appropriate
8. Conclusion and future work (1 slide)

10. Use of AI Tools

The use of AI tools is permitted but must be clearly stated and disclosed. Include an **Acknowledgement** section in the final report stating which tools you used and how and stating that the suggestions of these tools were reviewed carefully by the authors and their correctness validated. You remain fully responsible for everything you submit. Fabricated references or unverified results will be penalized. Please consult the course syllabus regarding rules for the use of AI in the course.

11. Evaluation

Projects shall be evaluated as per the table below.

Component	Weight
Project proposal	10%
Final report	40%
Code and reproducibility	20%
Presentation	20%
Individual contribution	10%
Total	**100%**

Note: in case a team member is reported to make no contribution to the project by other team members, the Individual contribution in the table above may be overridden by 100%

Final report grading criteria

Criterion	What is assessed
Problem and motivation	Clarity of the problem, relevance, and a precise statement of contributions
Technical quality	Correctness and appropriateness of the methods; for Track A, the suitability of the method for the problem; for Track B, the soundness of the idea; for Track C, the fidelity of the reproduction
Experimental rigour	Splits, baselines, metrics, variability, ablations, and absence of leakage
Analysis and insight	Error analysis, discussion of limitations, and the quality of conclusions
Presentation of the report	Organization, figures and tables, writing, and correct citations